

An even-strand Markov calculus for classical and virtual links

Alec Kriebel

ORCID: 0009-0001-9320-500X

Abstract

We give an explicit affirmative answer to the ordinary-closure interpretation of Problem 42 in the 2014 survey of Fenn, Ilyutko, Kauffman and Manturov. Four reversible algebraic schemes classify oriented classical link closures using only even-strand braid states. Four additional exchange schemes give the corresponding statement for oriented virtual links. The proof is an elementary conversion of the established unrestricted Markov theorems: positively stabilize each odd-strand vertex and lift every generating edge. Arbitrary finite word blocks are allowed; minimality and uniform geometric locality are not asserted. No historical-priority or present-openness claim is made.

1 Scope and conventions

Problem 42 of [1, p. 37], also on p. 34 of its author preprint, asks for a Markov formulation involving only even-strand braids. We retain *ordinary oriented, unframed closure* throughout. We supply both classical and virtual versions, since the item's braid terminology is unqualified. Plat closure, framing and transverse isotopy are different classifications. An even-strand plat theorem would not answer the ordinary-closure statement: the identity two-strand braid has a two-component ordinary closure and a one-component plat closure.

A state is a tagged word (N, w) , where $N \geq 2$ is even and all generator indices lie in $\{1, \dots, N - 1\}$. Tags cannot be suppressed: empty words on two and four strands close to different unlinks. Words multiply by concatenation; w^{-1} reverses the letters and inverts each one. Classical words use $\sigma_i^{\pm 1}$; virtual words additionally use $v_i = v_i^{-1}$. Write $W_N(k)$ for words whose indices are at most k , always regarded as words on the stated even N strands. The empty word is allowed, also for $k = 0$. These are syntactic support conditions, not tests of membership up to an unspecified braid equivalence. Let $s(w)$ increase every index by one, retaining the types and signs of letters. Put

$$G_i^{\text{cl}} = \{\sigma_i, \sigma_i^{-1}\}, \quad G_i^{\text{vir}} = \{\sigma_i, \sigma_i^{-1}, v_i\}.$$

Use the appropriate G_i below.

At every fixed even level we allow inverse cancellation and the Artin relations

$$\sigma_i \sigma_j = \sigma_j \sigma_i \quad (|i - j| > 1), \quad \sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}.$$

For virtual words we additionally allow exactly

$$\begin{aligned} v_i^2 &= 1, & v_i v_j &= v_j v_i \quad (|i - j| > 1), & v_i v_{i+1} v_i &= v_{i+1} v_i v_{i+1}, \\ \sigma_i v_j &= v_j \sigma_i \quad (|i - j| > 1), & \sigma_i v_{i+1} v_i & & &= v_{i+1} v_i \sigma_{i+1}. \end{aligned}$$

All these relations may be used in arbitrary word contexts, in either direction. No welded forbidden relation is imposed.

2 The even-endpoint schemes

All following arrows are reversible and have only even-strand endpoints. The four common schemes are

$$(N, b) \longleftrightarrow (N, aba^{-1}), \quad a, b \in W_N(N-1), \quad (C)$$

$$(N, b\sigma_{N-1}) \longleftrightarrow (N, aba^{-1}\sigma_{N-1}), \quad a, b \in W_N(N-2), \quad (BC)$$

$$(N, b\sigma_{N-1}) \longleftrightarrow (N, bg), \quad b \in W_N(N-2), \quad g \in G_{N-1}, \quad (T)$$

$$(N, b) \longleftrightarrow (N+2, bg\sigma_{N+1}), \quad b \in W_N(N-1), \quad g \in G_N. \quad (D)$$

In (D), the old word is placed on the first N strands of the new $N+2$ -strand state. The identity choice in (T) may be omitted. These four schemes, with classical letters only, form the classical calculus.

The virtual calculus includes four further schemes. For $a, b \in W_N(N-2)$, allow

$$(N, a\sigma_{N-1}^{-1}b\sigma_{N-1}) \longleftrightarrow (N, av_{N-1}bv_{N-1}), \quad (R)$$

$$(N, s(a)\sigma_1^{-1}s(b)\sigma_1) \longleftrightarrow (N, s(a)v_1s(b)v_1). \quad (L)$$

For even $N \geq 4$ and $a, b \in W_N(N-3)$, allow

$$(N, a\sigma_{N-2}^{-1}b\sigma_{N-2}\sigma_{N-1}) \longleftrightarrow (N, av_{N-2}bv_{N-2}\sigma_{N-1}), \quad (BR)$$

$$(N, s(a)\sigma_1^{-1}s(b)\sigma_1\sigma_{N-1}) \longleftrightarrow (N, s(a)v_1s(b)v_1\sigma_{N-1}). \quad (BL)$$

In (BL), the shifted blocks have indices at most $N-2$; the separate retained terminal crossing has index $N-1$ and is not shifted. There are four classical or eight virtual *scheme families*, apart from the defining relations. Blocks may have arbitrary finite length. Neither a minimal list nor a bound on the number of strands in a geometric support disk is claimed.

Theorem 1. *Every nonempty oriented classical link is the ordinary closure of an even-strand classical braid. Two such tagged braids have isotopic oriented closures if and only if they are connected by defining relations at even levels and the schemes (C), (BC), (T), (D), with classical letters only.*

Every nonempty oriented virtual link has an even-strand virtual braid representative. Two such tagged virtual braids have equivalent oriented virtual closures if and only if they are connected by defining virtual braid relations at even levels and (C), (BC), (T), (D), (R), (L), (BR), (BL).

3 Proof by positive padding

The classical inputs are Alexander's representation theorem and ordinary Markov equivalence by conjugation and signed right stabilization; see [2, Theorems 2.1–2.2]. The virtual inputs are Kamada's Proposition 3.1 and Theorem 3.2 in [3]. In addition to relations and conjugation, they permit right stabilization by $\sigma_m^{\pm 1}$ or v_m , and both exchanges at total count m :

$$a\sigma_{m-1}^{-1}b\sigma_{m-1} \longleftrightarrow av_{m-1}bv_{m-1}, \quad s(a)\sigma_1^{-1}s(b)\sigma_1 \longleftrightarrow s(a)v_1s(b)v_1, \quad (1)$$

where the unshifted blocks use indices at most $m-2$. The signs and shift are also stated in [4, §5, p. 30]. These unrestricted theorems are imported, not reproved here.

Proof of Theorem 1. First verify soundness without using completeness. (C) is conjugation. For (BC), the two prefixes on $N - 1$ strands are conjugate, and both endpoints are their positive right stabilizations. (T) compares two permitted right stabilizations of the same prefix on $N - 1$ strands. (D) is a permitted stabilization followed by a positive stabilization. (R) and (L) are (1); in (BR) and (BL), apply that exchange on $N - 1$ strands and then positively stabilize both sides. The support bounds make each auxiliary operation legitimate. Thus every listed replacement preserves ordinary oriented closure. Odd-strand objects used to verify this fact are auxiliary proof objects, not allowed states of the displayed calculus.

For completeness, define, on unrestricted tagged words with $m \geq 1$,

$$P(m, w) = \begin{cases} (m, w), & m \text{ even}, \\ (m + 1, w\sigma_m), & m \text{ odd}. \end{cases} \quad (2)$$

Its closure agrees with the original closure by positive stabilization, and it fixes every even tagged word exactly. Given an unrestricted Markov chain between even endpoints, apply P to every vertex. We check its edges exhaustively.

A defining relation at even count remains unchanged. At odd count the same relation is applied inside its entire old word context, with the positive tail retained. Right inclusion respects every defining relation; hence no group-word ambiguity is hidden by padding.

Conjugation $b \longleftrightarrow aba^{-1}$ at even count m becomes (C). At odd m it becomes (BC) at $N = m + 1$, since both old blocks have indices at most $m - 1 = N - 2$. For stabilization $(m, b) \longleftrightarrow (m + 1, bg)$, $g \in G_m$, the two images are

$$\begin{cases} (m, b) \longleftrightarrow (m + 2, bg\sigma_{m+1}), & m \text{ even} : (D), \\ (m + 1, b\sigma_m) \longleftrightarrow (m + 1, bg), & m \text{ odd} : (T). \end{cases}$$

This includes both classical signs and virtual stabilization. Reversed arrows also cover every destabilization.

For a right exchange at even total count m , the images give (R) at $N = m$. At odd total count m , its exchange index is $m - 1 = N - 2$, its unshifted blocks have indices at most $m - 2 = N - 3$, and padding appends $\sigma_m = \sigma_{N-1}$: this is (BR). A left exchange similarly gives (L) at even m and (BL) at odd m . In the latter, only the old blocks are shifted, so their indices are at most $N - 2$ and the padding crossing remains separate. The first possible odd exchange count is $m = 3$, explaining $N \geq 4$ for (BR) and (BL).

This covers every unrestricted edge. Delete identical successive images if desired. The resulting chain has only even states and the original endpoints. The classical chain needs no virtual letters or exchanges. Soundness proves the converse, and the corresponding Alexander theorem followed by P proves the representation statements. $\square$

Corollary 2. *An unrestricted certificate with maximum strand count M converts to an even certificate with maximum strand count $2\lceil M/2 \rceil$.*

Proof. Each vertex count m becomes $m + (m \bmod 2)$, and every lifted edge above has exactly its displayed endpoints. This function is nondecreasing in m . $\square$

This is a bound for a *supplied* certificate, not an algorithm or complexity bound for finding one. At one strand the empty word pads to $(2, \sigma_1)$, the unknot. At $N = 2$, (BC)'s blocks are empty and the virtual exchanges are identities. If the empty link is desired, adjoin an isolated zero-strand empty state; no positive-strand ordinary closure is empty.

4 Double moves and checkable support boundaries

Fiedler [5] reports that conjugation and double Markov moves alone do not replace ordinary Markov equivalence. Our calculus includes the additional (BC) and (T) schemes. Indeed, (D) followed by (T) at $N + 2$ gives every classical double stabilization

$$(N, b) \longleftrightarrow (N + 2, b\sigma_N^\varepsilon\sigma_{N+1}^\delta), \quad \varepsilon, \delta \in \{+1, -1\},$$

because the prefix $b\sigma_N^\varepsilon$ has indices at most N . This does not show that (BC) or (T) is generated by conjugation and double moves. Our comparison uses Fiedler’s publisher abstract and identified publisher first-page preview; we do not independently certify his full counterexample proof. The elementary padding derivation establishes the stated formulation without a historical-priority claim or an assertion that the 2014 question remains open today.

The block restrictions have mathematical content. If (T) at $N = 2$ were allowed with $b = \sigma_1^2$, it would relate σ_1^3 to σ_1 after cancellation. Their closures both have one component, but their Fox 3-coloring counts are 9 and 3. With the positive generator drawn with its left strand over the right, the color map is $R(x, y) = (2x - y, x)$ over $\mathbb{F}_3$; $R^3 = 1$ and R fixes precisely $x = y$. The invertible coloring rule gives bijections under the Reidemeister moves, so these counts rule out that enlarged scheme. The true (T) requires an empty prefix here.

There is a virtual obstruction missed by component counts as well. Enlarging (R) at $N = 2$ to allow $a = b = \sigma_1$ would relate σ_1^2 to $\sigma_1 v_1 \sigma_1 v_1$. Their signed ordered intercomponent crossing matrices are respectively $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $\begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}$, distinct even after simultaneous component relabeling. These matrices are invariant under oriented virtual Reidemeister moves: RI adds only self-crossings, classical RII cancels signs, RIII preserves each ordered component pair, and virtual detours introduce no classical crossings. The genuine (R) excludes these blocks.

The accompanying standard-library Python diagnostics check explicit lifted endpoints, every relation family in bounded word contexts, support and tags, and exact countercontrols such as these. Finite diagnostics supplement the universal proof; they are not a link-equivalence oracle, a braid word-problem solver or formal verification of the imported theorems.

AI and review disclosure. AI tools were used extensively in constructing and drafting this formulation, and in adversarial review and computational verification. This note is unrefereed; no human peer review or formal proof certification is claimed.

References

- [1] R. Fenn, D. P. Ilyutko, L. H. Kauffman and V. O. Manturov, *Unsolved problems in virtual knot theory and combinatorial knot theory*, Banach Center Publications **103** (2014), 9–61. doi:10.4064/bc103-0-1; arXiv:1409.2823v1.
- [2] E. Gorsky, O. Kivinen and J. Simental, *Algebra and geometry of link homology: Lecture notes from the IHES 2021 Summer School*, Bull. London Math. Soc. **55** (2023), no. 2, 537–591. doi:10.1112/blms.12761; arXiv:2108.10356v1.
- [3] S. Kamada, *Braid presentation of virtual knots and welded knots*, arXiv:math/0008092v1 (2000), §§2–3, Proposition 3.1 and Theorem 3.2.
- [4] L. H. Kauffman and S. Lambropoulou, *Virtual braids and the L-move*, J. Knot Theory Ramifications **15** (2006), 773–811; arXiv:math/0507035v3, §5, p. 30.
- [5] T. Fiedler, *Markov moves cannot be replaced by double Markov moves*, J. Knot Theory Ramifications **12** (2003), no. 4, 575–577. doi:10.1142/S0218216503002561.